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Vertical contraction scour beneath solid and porous obstacles in steady currents: A numerical and theoretical study

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ABSTRACT

Vertical contraction scour occurs when the flow is vertically obstructed by submerged structures, resulting in increased shear stress and scour on the sand bed. In the present study, vertical contraction scour under solid and porous obstacles is investigated with a fully coupled hydrodynamic and morphological sediment transport model and theoretical analysis. The numerical model is validated against our experiments of scour below a rectangular slab. We examined the influence of distance between submerged obstacles and the undisturbed bed (i.e., obstacle elevation) on scour depth. In addition, the effects of obstacle height, length and porosity on scour depth are investigated. A general relationship between obstacle elevation and scour depth is identified, showing three regimes: very small elevation, intermediate elevation, and high elevation when the obstacle emerges above the water surface. A maximum scour depth is observed when the obstacle elevation is approximately 0.15 times the flow depth. Increasing porosity significantly reduces vertical contraction scour, and when porosity exceeds 75 %, the scour depth becomes negligible. The theoretical analysis conducted in the present study demonstrates its ability to estimate the flow rate distribution above and below the obstacles, along with its corresponding effects on bed shear stress. Furthermore, an empirical equation is proposed for predicting pressure scour beneath porous obstacles, showing acceptable accuracy for engineering applications.

1. Introduction

Vertical contraction scour, also known as pressure scour, is widespread and causes a significant threat to the safety of hydraulic and coastal structures. As shown in Fig. 1, vertical contractions can occur in various forms in riverine and coastal environments, including bridge inundations, debris accumulation, floating vegetated wetland, floating breakwaters and certain types of wave energy converters.

Bridge inundation is one of the most common causes of vertical contraction scour, occurring when the bridge deck is submerged during flash flood, storm surge, or tsunami events. As climate change intensifies the extreme weather conditions, the risk of bridge inundation is expected to rise, making the bridges even more vulnerable to inundation-induced damages. Most of the existing studies aimed for predictive equations of the equilibrium scour depth beneath the submerged bridge deck (Umbrell et al., 1998; Arneson and Abt, 1998; Melville, 2014), providing stability and safety guidelines for the bridge design. However, the underlying mechanisms of vertical contraction scour beneath solid obstacles, especially the influence of structural characteristics, such as the obstacle elevation and geometric dimensions on the flow rate distribution, bed shear stress and the corresponding impact on local scour are not fully understood. Therefore, theoretical, numerical, and experimental approaches are used in the present study to further address these gaps.

Porous obstacles can also induce vertical contraction scour. For example, floating debris, comprised of woody branches, logs, soil, and even rocks, is commonly transported by flow during extreme hydrodynamic conditions such as floods and tsunami events. These materials often accumulate at natural or artificial obstructions in both riverine and coastal zones, like narrow channels, bridge piers (Melville and Dongol, 1992), and beaches (Grilliot et al., 2019), potentially obstructing water flow and leading to an increased scour due to the pressured flow. Spanwise large wood accumulation was experimentally investigated by Schalko et al. (2019a, 2019b), and its effect on back water rise and local scour was discussed. With spanwise large wood accumulation, the maximum scour depth was observed to be higher than the approaching flow depth. Other studies, including Pagliara and Carnacina (2011), Ebrahimi et al. (2018), Panici and de Almeida (2020) have investigated

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the effects of woody debris induced vertical contraction on bridge pier scour. Laboratory experiments conducted by Gui et al. (2024) tested the influence of floating vegetation on turbulence and sediment deposition. The results revealed that the presence of floating vegetation increased flow velocity and intensified turbulent kinetic energy (TKE) beneath the floating vegetation cluster. This hydrodynamic reduced sediment deposition, which, in turn, increased the potential risk of sand bed erosion due to enhanced shear stress. In addition, floating fishing nets used in offshore aquaculture can modify the surrounding flow field and induce potential seabed erosion (Ji et al., 2025; Fan et al., 2024). Compared to other structures, accumulated woody debris, floating vegetated wetland, and fishing nets exhibit a certain degree of porosity.

Existing studies mainly focused on vertical contraction scour caused by solid obstacles, the scour induced by porous obstacles remain underexplored. Particularly, how different porosity levels affect the vertical contraction scour depth have not been well studied yet. To address these gaps, numerical simulations and theoretical analysis are conducted in the present study to investigate vertical contraction scour below porous obstacles under steady current conditions.

The aim of this study is to enhance understanding of how obstacle elevation, geometric dimension and porosity influence velocity distribution, and their subsequent effects on scour depth. Idealized rectangular models are employed as a representative of the obstacle, and the porosity of submerged structures is analyzed to assess the effects of porous obstacles, such as debris accumulation and floating vegetated wetland on scour process.

2. Methodology

2.1. Numerical model set-ups

The numerical investigation is conducted in an open-source toolbox, OpenFOAM®, foam-extend-3.1. A fully-coupled hydrodynamic and morphological model (*sediMorph*) (Jacobsen, 2011; Jacobsen et al., 2014; Jacobsen and Fredsøe, 2014) is used in the present study.

2.1.1. Hydrodynamic model

The hydrodynamic model is based on solving incompressible Reynold-averaged-Navier-Stokes (RANS) equations with a k-ω closure, including the continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{1}$$

and the momentum equation:

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(2\nu S_{ij} + \frac{\tau_{ij}}{\rho}\right) \tag{2}$$

where u_i are the mean velocities; x_i are the Cartesian coordinates; p is the pressure; t is the time; ν is the kinematic viscosity of the fluid; ρ is the density of the fluid, and:

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{3}$$

$$\frac{\tau_{ij}}{\rho} = -\overline{u_i' u_j'} = 2\nu_T S_{ij} - \frac{2}{3}k\delta_{ij} \tag{4}$$

where S_{ij} is the mean strain rate tensor; τ_{ij} is the Reynolds stress tensor; δ_{ij} is the Kronecker delta; ν_T is the eddy viscosity. The turbulent kinetic energy density per unit mass is:

$$k = \frac{1}{2}\overline{u_i' u_i'} \tag{5}$$

The two-equation k-ω closure is applied in this model (Wilcox, 2006), where the eddy viscosity is defined as:

$$\nu_T = \frac{k}{\widetilde{\omega}}, \widetilde{\omega} = \max\left\{\omega, C_{\lim}\sqrt{\frac{2S_{ij}S_{ij}}{\beta^*}}\right\} \tag{6}$$

where $C_{\lim} = 7/8$. The two-equation k-ω closure includes a turbulent kinetic energy (k) equation:

$$\frac{\partial k}{\partial t} + u_j\frac{\partial k}{\partial x_j} = \frac{\tau_{ij}}{\rho}\frac{\partial u_i}{\partial x_j} - \beta^* k\omega + \frac{\partial}{\partial x_j}\left[\left(\nu + \sigma^*\frac{k}{\omega}\right)\frac{\partial k}{\partial x_j}\right] \tag{7}$$

and a specific dissipation rate (ω) equation:

$$\frac{\partial \omega}{\partial t} + u_j\frac{\partial \omega}{\partial x_j} = \alpha\frac{\omega}{k}\frac{\tau_{ij}}{\rho}\frac{\partial u_i}{\partial x_j} - \beta\omega^2 + \frac{\sigma_d}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j} + \frac{\partial}{\partial x_j}\left[\left(\nu + \sigma\frac{k}{\omega}\right)\frac{\partial \omega}{\partial x_j}\right] \tag{8}$$

$$\sigma_d = H\left\{\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}\right\}\sigma_{d0} \tag{9}$$

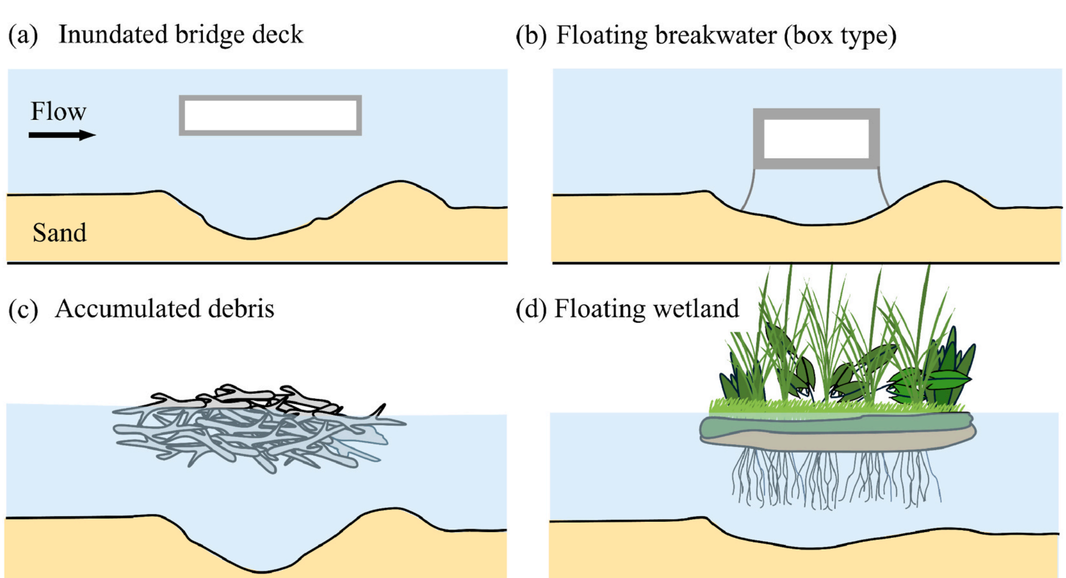

Fig. 1. Schematic drawings of vertical contraction scour beneath: (a) an inundated bridge deck; (b) a floating breakwater; (c) accumulated debris; and (d) a floating wetland.

where $\beta^* = 0.09$, $\sigma^* = 0.6$, $\alpha = 0.52$, $\beta = 0.0708$, $\sigma = 0.5$, $\sigma_{d0} = 0.125$; and $H\{\cdot\}$ is the Heaviside step function. $H\{x\} = 1$ when x is positive, otherwise $H\{x\} = 0$.

2.1.2. Sediment transport and morphological model

The sediment transport model, consisting bedload and suspended sediment transport models, was proposed and evaluated in details by Engelund and Fredsøe (1976), Fredsøe and Deigaard (1992), Roulund et al. (2005) and Jacobsen (2011).

The bedload transport rate q_b is calculated as:

$$q_b = \frac{\pi}{6} d^3 \frac{P_{EF}}{d^2} U_b \tag{10}$$

where d is the median diameter of sediment particles. To keep consistent with our experiments for validation, $d = 0.84$ mm is used in the present study; U_b is the bed load particle moving velocity, which is determined as:

$$U_b = a' U_f \left(1 - 0.7\sqrt{\frac{\theta_{c0}}{\theta}\left(\cos(\gamma) - \frac{1}{\mu_d}\sin(\gamma)\right)}\right) \tag{11}$$

where U_f is the friction velocity, γ is the slope angle, $a' \approx 10$ is a non-dimensional coefficient, P_{EF} is the percentage of particles in motion in the surface layer of the bed,

$$P_{EF} = \left[1 + \left(\frac{\frac{1}{6}\pi\mu_d}{\theta - \theta_c}\right)^4\right]^{-1/4} \tag{12}$$

where the dynamic friction coefficient $\mu_d = 0.70$ is used in the present study. The Shields parameter θ is defined as:

$$\theta = \frac{U_f^2}{(s-1)gd} \tag{13}$$

and the critical Sheilds parameter θ_c on a sloped bed is calculated as:

$$\theta_c = \theta_{c0}\left(\cos(\gamma) - \frac{1}{\mu_s}\sin(\gamma)\right) \tag{14}$$

where s is the specific gravity of sediment, g is the gravity acceleration, the critical Shields parameter for a horizontal bed θ_{c0} is taken as 0.04, and the static friction coefficient μ_s is taken as 0.65 in the present study.

The suspended load transport model solves the following equation:

$$\frac{\partial c}{\partial t} + (u_j - w_s\delta_{j3})\frac{\partial c}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \beta_s\frac{k}{\omega}\right)\frac{\partial c}{\partial x_j}\right] \tag{15}$$

where c is the concentration of the suspended sediment and w_s is the sediment settling velocity, based on the grain size d. In the present study, β_s is set to 1, assuming that the sediment diffusivity is equal to the eddy viscosity. This equation is solved on a sub-set mesh where near bed cells below a reference level are removed, as described in detailed by Jacobsen et al. (2014). The concentration at the reference level (c_b) is calculated as:

$$c_b = \frac{c_0}{\left(1 + \frac{1}{\lambda_b}\right)^3} \tag{16}$$

where $c_0 = 0.6$, and the linear concentration λ_b is:

$$\lambda_b^2 = \frac{\kappa^2\alpha_1^2}{0.013 s\theta}\left(\theta - \theta_c - \frac{1}{6}\pi\mu_d P_{EF}\right) \tag{17}$$

A reference level $b = 3.5d$ is utilized in the present study following Fuhrman et al. (2014).

For bed morphology, the bed elevation h_b can be calculated by a sediment continuity (Exner) equation:

$$\frac{\partial h_b}{\partial t} = \frac{1}{1 - p_b}\left[-\frac{\partial q_{bi}}{\partial x_i} + D_s + E_s\right], i = 1, 2 \tag{18}$$

where p_b is the bed porosity ($p_b = 0.4$ in the present study), and the deposition (D_s) and erosion (E_s) stemming from the suspended model can be calculated as:

$$D_s = (w_s - u_3)c_b, E_s = (\nu + \beta_s\nu_T)\left.\frac{\partial c}{\partial x_3}\right|_{x_3 = b} \tag{19}$$

In the present model, the morphological time step and the hydrodynamic time step are equivalent. To ensure that the bed slope angle does not exceed the underwater angle of repose of the sediment, a physically based sand slide model (Niemann et al., 2011) is used in the present study. More details of the *SediMorph* model used in the present study can be found in Jacobsen et al. (2014).

2.1.3. Porous zone setup

The porous zone is determined by the Darcy-Forchheimer method, which is a widely used model for describing fluid passing through porous media. the Darcy-Forchheimer equation acts as a sink term in the momentum equation:

$$\frac{\partial}{\partial t}(\varphi u_i) + u_j\frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[2\nu S_{ij} + \frac{\tau_{ij}}{\rho}\right] + \frac{1}{\rho}S_i \tag{20}$$

where the value of φ should be between 0 and 1, and the latter is a complete porosity. S_i is composed of two parts, a viscous loss term and an inertial loss term, creating a pressure drop that is proportional to the velocity and velocity squared, respectively. S_i is described by the Darcy-Forchheimer equation:

$$S_i = -\left(\mu D_{ij} + \frac{1}{2}\rho|u_{kk}|F_{ij}\right)u_i \tag{21}$$

where S_i can be expressed as a pressure gradient, for the two-dimensional, homogenous porous media with streamwise flow, we can write Eq. (21) as follows:

$$\nabla p = \frac{\Delta p}{\Delta x} = \mu D u + \frac{1}{2}\rho F u^2 \tag{22}$$

where D and F are the Darcy coefficient and Forchheimer coefficients, respectively. These two parameters are set as input parameters for the simulation of different porosities. In the present study, the porous obstacle is idealized similar as a thick perforated plates (Johann, 2010) or an array of cylinders (Spröer et al., 2025), which leaves out the linear (viscous) terms (make D value = 0) and simplify the pressure drop to a quadratic problem, and assuming the porosities at all directions are the same. Empirical equations (Johann, 2010) are applied to determine the Forchheimer coefficients F as follows:

$$\alpha_d = 0.6 + 0.4\varphi^2 \tag{23}$$

$$k_r = \left(\frac{1}{\alpha_d} - 1\right)^2 + (1 - \varphi)^2 \tag{24}$$

$$F = \frac{k_r}{\varphi^2 L} \tag{25}$$

where α_d is discharge coefficient, k_r is resistance coefficient, φ is the porosity value (open volume to total volume ratio), L is the length of the obstacle.

2.1.4. Boundary conditions

The free surface is not considered in the present study, and the top of the computational domain is set as a frictionless slip wall. This “lid” effect has been discussed by Roulund et al. (2005) and Baykal et al. (2015) in pile scour, suggesting that this model is applicable for simulating the conditions without significant surface variations, e.g., scour under steady current. With an additional body force, a frictionless slip wall at the top boundary is adopted to simulate scour under both wave only and combined wave-current conditions (Fuhrman et al., 2014; Li et al., 2020b). The sand bed is modeled as a rough wall, and a generalized wall function is employed to represent the influence of wall roughness. The friction velocity is determined from the tangential velocity at the nearest cell center based on an assumed logarithmic velocity distribution:

$$\frac{u}{U_f} = \frac{1}{\kappa}\ln\frac{30y_c}{k_N} \tag{26}$$

$$k = \frac{U_f^2}{\sqrt{\beta^*}} \tag{27}$$

$$\omega = \frac{U_f}{\sqrt{\beta^*}\kappa\Delta y} \tag{28}$$

where $y_c = \Delta y/2$ is the normal distance from the wall to the cell center, with Δy being the cell thickness, $\kappa = 0.4$ is the von Karman constant, and k_N is Nikuradse's equivalent sand roughness, $k_N = 1.0d$ is used in this study rather than the widely used $k_N = 2.5d$. This is because our experiments have a dense and well levelled sand bed (Schlichting and Gersten, 2003), and the selected value can provide a more reasonable results compared to the experiments. Similar selections can also be found in the studies by Fuhrman et al. (2010) and Larsen et al. (2020), where $k_N = 1.4d$ was used.

The surface of the solid obstacles are modeled as smooth wall, and the friction velocity is based on Cebeci and Chang (1978), which is also introduced by Fuhrman et al. (2014). The generalized wall functions for k and ω are defined following Fuhrman et al. (2014):

$$\frac{\kappa}{U_f^2} = \min\left\{A\Delta y^{+2}, \frac{1}{\sqrt{\beta^*}}\right\} \tag{29}$$

$$\frac{\omega\nu}{U_f^2} = \max\left\{\frac{B}{\Delta y^{+2}}, \frac{1}{\sqrt{\beta^*}\kappa\Delta y^{+}}\right\} \tag{30}$$

where $\Delta y^+ = \Delta y U_f/\nu$, $A = 0.02466$, and $B = 96.885$.

A two-dimensional mesh is adopted, with local refinement near the bed and the structure. We have also examined three different mesh resolutions near the bed, with the first mesh size about 0.5d, 1.0d, and 1.5d. Insignificant variations in scour depth were observed. A medium bottom mesh resolution is then used, the resulting y^+ value is around 50, which falls well within the recommended range. To reduce the domain size and improve computational efficiency, a Dirichlet boundary condition with u, k and ω from a preliminary simulation is applied to the inlet boundary, the outlet boundary condition is set as zero gradient of pressure, and an *inletOutlet* condition is applied for velocity, allowing fluid to exit freely. The final computational domain extends to about ten times the structure length and the final mesh contains approximately 17000 cells. Although the dynamic mesh approach captures the bed morphology with good accuracy, it is computationally demanding; each simulation requires about 10 days in parallel computation on 8 cores to reach a physical time of 7200 s.

2.2. Experimental model set-ups

The capability and stability of the *sediMorph* model have been demonstrated through its wide use for scour simulations under various conditions, such as scour around monopile foundations under current and waves (Baykal et al., 2015; Baykal et al., 2017; Li et al., 2021), tsunami-induced scour around monopile (Larsen et al., 2017; Hu et al., 2025; Hu and Li, 2025), and scour beneath submarine pipelines (Fuhrman et al., 2014; Li et al., 2020a, 2020b).

In the present study, to further validate the model on vertical contraction scour, experiments are conducted with two obstacle elevations, corresponding to e/y values of 0.225 and 0.6. The experiments are conducted in a 440 mm wide water-sand recirculating flume at the University of Auckland. A rectangular slab, with a dimension of 0.23m × 0.44m × 0.055m ($l \times W \times h$) is used as the model of solid obstacle, where h, l and W are the height, longitudinal length, and lateral width of the obstacle, respectively. More information about the flume can be found in Xie et al. (2022) and Tang et al. (2025). The sediment used in the experiment has a median size d_{50} of 0.84 mm and a standard deviation σ_g equal to 1.3, with $d_{16} = 0.65$ mm, $d_{30} = 0.73$ mm, $d_{84} = 1.10$ mm, and $d_{90} = 1.15$ mm. The same sediment size is adopted in the numerical model. The depth-averaged approaching velocity u_0 is 0.355 m/s, the critical mean flow velocity u_c is 0.382 m/s, leading to a near threshold clear water condition with $u_0/u_c = 0.93$ both in the experimental and numerical studies, where u_c is determined by sediment properties (Melville and Coleman, 2000). Fig. 2 shows the key variables of the experiments and the present numerical simulations, where e is the elevation of the obstacle above the bed, w is the distance between the obstacle top to the water surface, y denotes the flow depth, which is maintained at a constant value of 0.2 m among all the experiments and simulations, and S is the scour depth. The scale between numerical and experimental study is 1:1.

3. Model validation

Figs. 3 and 4 illustrates the experimental and numerical results of temporal scour evolution and scour profile with vertical contractions under various initial elevations, and the non-dimensional time t^* is defined as:

$$t^* = \frac{\sqrt{g(s-1)d^3}}{h^2}t \tag{31}$$

where t is the physical time. In the current study, the physical time t is set as 7200 s (the corresponding t^* is 233.13) when the scour depth has passed transition phase and reached at equilibrium stage (Ettema, 1980).

The numerical results of temporal scour evolution below the middle of the obstacle are compared with the experimental results in Fig. 3. Both results of the cases with e/y values equal to 0.225 and 0.6 present good agreement with the experimental data, except a slightly higher scour acceleration at the initial stage. The velocity profile extracted from the case without obstacles, as shown in Fig. 3(c), also demonstrates good agreement with the experimental data and confirms that the model accuracy is sufficient.

Fig. 4 shows a comparison of the scour profiles between the numerical results and the experimental data. In Fig. 4(a), the predicted scour depth for the case of $e/y = 0.225$ is slightly higher than the measured one below the downstream side of the rectangular slab. However, the scour profiles and the temporal scour evolution at the end of the test compare well with the experimental data. The strong agreement between the numerical and experimental results demonstrates the validity of the present model and its potential for further applications.

In the present study, an idealized model, e.g., a rectangular slab, is used as the model of solid and porous obstacles. The effect of obstacle elevation (e/y) is studied in this study. In addition, the influence of structural dimensions, including obstacle height (h/y) and length (l/y), is also investigated. Porosity (φ) is included for simulating vertical contraction scour induced by porous obstacles. The flow depth $y = 0.2$ m and sediment median size $d = 0.84$ mm are also constant among the tests.

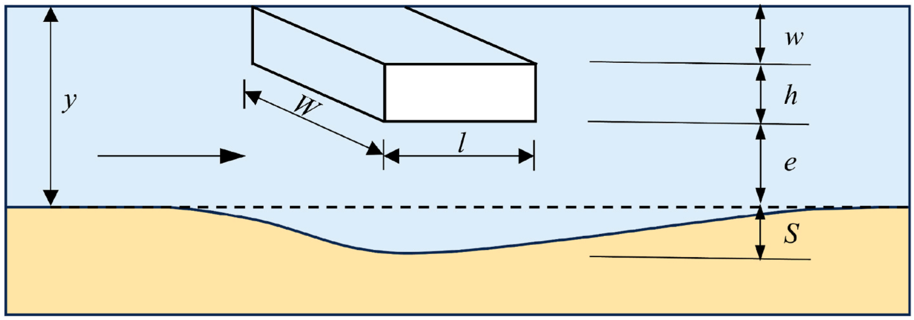

Fig. 2. Sketch of vertical contraction scour.

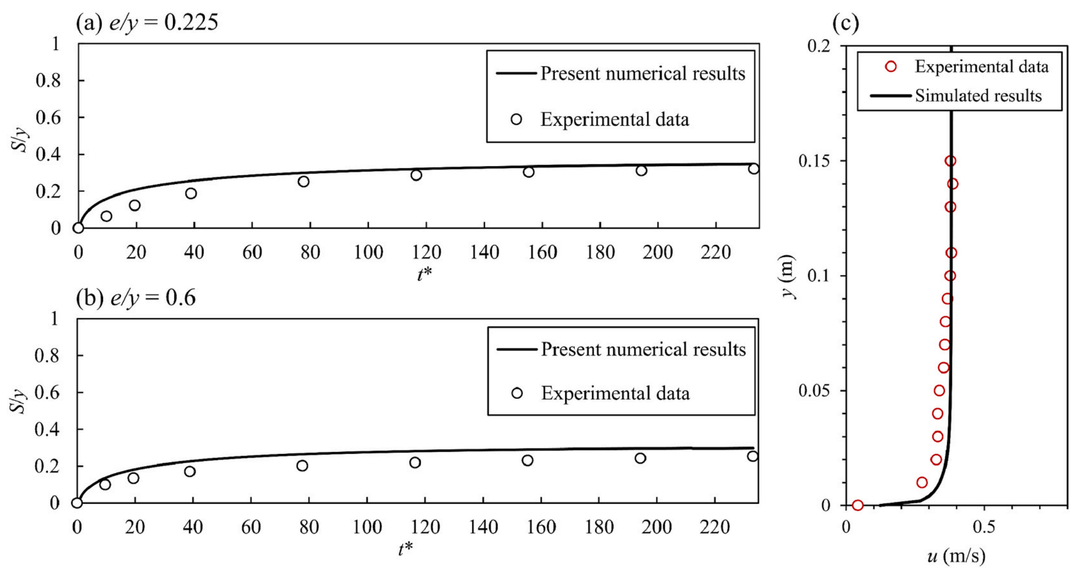

Fig. 3. Comparison between the numerical and experimental results of: (a) temporal scour evolution at $e/y = 0.225$; (b) temporal scour evolution at $e/y = 0.6$; and (c) velocity profile without obstacles.

4. Numerical simulations of scour beneath solid obstacles

Scour beneath solid obstacles are numerically studied in this section. As listed in Table 1, there are totally ten cases with varying obstacle elevations, the e/y values located in the range of 0.05 and 0.9, and the obstacle height and length remain constant among all the cases.

To further address the influence of obstacle height and length on the scour profile and maximum scour depth, additional simulations with systematically varied obstacle heights and lengths are conducted. When selecting the varying h/y and l/y values, we aim to avoid geometries that have unrealistically length to height ratio. This allows the adopted structures to remain representative of real and practical configurations.

As listed in Table 2, ten additional cases are considered, comprising four with reduced height, three with increased height, and three with reduced length. The case numbers correspond to those in Table 1, ensuring that the flow depth and obstacle elevation remain consistent for cases with the same number. Subscripts a, b, and c denote reduced obstacle height, increased obstacle height, and reduced obstacle length, respectively.

4.1. Scour profile beneath solid obstacle with varying obstacle dimensions

Scour profiles for four typical cases with different obstacle elevations are shown in Fig. 5. The dotted, dashed, dash-dot, and solid lines represent four non-dimensional times: $t^* = 1.94$, 19.43, 116.57, and 233.13, respectively. Fig. 5(a)–(c) display fully submerged obstacles with low and moderate e/y values, while Fig. 5(d) shows a partially submerged case with high e/y values. For the low e/y value of 0.05, a steep upstream scour slope is observed. At moderate e/y ratio, the scour profile becomes smoother, and the upstream slope is less significant. As the e/y value increases, both the maximum scour depth and the upstream slope decrease. This is intuitive as increasing e/y results in less pressured flow beneath the solid obstacle. Additionally, scour profiles in all four cases show insignificant evolutions between $t^* = 116.57$ and 233.13, indicating that the scour process is approaching an equilibrium stage. The maximum scour depth at each time t^* is indicated by a solid line with circular markers. For low e/y values, the maximum scour depth tends to occur beneath the upstream edge of the obstacle. As the e/y values increases, the location of the maximum scour depth shifts downstream to the center of the obstacle.

Based on Figs. 5(c) and 6 illustrates the influence of varying obstacle height and length on the scour profile under the conditions of $e/y = 0.325$. As shown in Fig. 6(a), with reduced obstacle height, scour depth is significantly reduced, and the scour profile tend to be smooth and flat. In Fig. 6(b), increasing obstacle height amplifies the scour depth, resulting in steeper scour hole and downstream dune height. This occurs because at a fixed obstacle elevation, greater obstacle height leads to increased blockage of the approaching flow, thereby enhancing the flow rate and bed shear stress beneath the obstacle. Fig. 6(c) demonstrates the influence of decreasing obstacle length. The results reveals that by changing obstacles length, the scour depth may not vary a lot as the blockage ratio to the approaching flow is not changed. However, a reduced obstacle length may result in maximum scour location moving upstream and alter the scour profile accordingly.

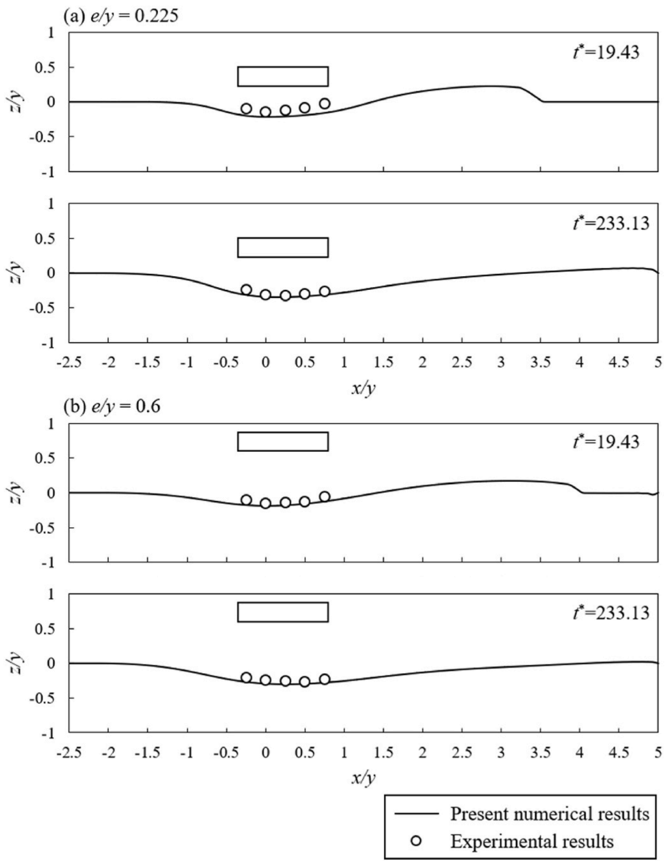

Fig. 4. Model validation of scour profile evolution between the experimental and the numerical results: (a) $e/y = 0.225$; (b) $e/y = 0.6$.

In general, the location of maximum scour beneath solid obstacles shifts downstream as the scour develops. By the end of the simulation, the maximum scour depth is near the obstacle centerline, with only a negligible rate of downstream movement.

4.2. Maximum scour depth beneath solid obstacles

Fig. 7 illustrates the maximum scour depth at the end of each simulation. The curve with circular markers and orange shadow represents the results with varying e/y values but constant obstacle heights and lengths, as the cases listed in Table 1. Generally, the scour depth decreases with the increase of e/y values, except the near bed range. As shown in Fig. 7, when the obstacle elevation e/y is small and the solid obstacle is close to the bed, an increasing trend is observed as e/y rises from 0.05 to 0.15. The underlying reasons will be discussed later in Fig. 8. It is also observed that when $e/y > 0.725$, the obstacle becomes partially submerged and the entire approaching flow passes beneath it. Under partially submerged conditions, the discharge below the obstacle remains constant, while further increases in obstacle elevation reduce the average flow velocity and shear stress beneath it. Consequently, the scour depth decreases more rapidly with increasing e/y than under fully submerged conditions.

Additionally, Fig. 7 includes the scour depths obtained for varying obstacle heights and lengths, as the cases listed in Table 2. Dashed lines with rectangles and diamonds represent simulations with decreased and increased obstacle heights (h/y), respectively. While triangles denote cases with various obstacle lengths (l/y). Increasing obstacle height at a fixed e/y value leads to greater scour depth. In contrast, changes in obstacle length have only a minor influence on maximum scour depth as mentioned before.

In the present study, the flow depth is constant, altering the obstacle height can shift the regime from fully to partially submerged. For instance, raising the obstacle height causes partial submergence at a lower elevation (e/y). Once the obstacle is already partially submerged, further increases in height above the water surface do not affect scour depth. As shown in Fig. 7, the three arrows indicate the transition points where the obstacle becomes partially submerged. Beyond these points, the scour depth curves will gradually overlap.

Moreover, Fig. 7 compares the present numerical results with the existing literature data from the experimental studies of Guo et al. (2009), Melville (2014) and Koushki et al. (2023). The current numerical results has a good agreement with the experimental data of Melville (2014) due to the very similar experimental set-up, including the same dimensions of the rectangular slab and similar flow intensities. Guo et al. (2009) tested two different obstacle height values, $h/y = 0.16$ and $h/y = 0.20$, both of which are smaller than those used in the varying e/y cases ($h/y = 0.275$), but larger than that in the reduced obstacle height cases ($h/y = 0.15$) in the present study. Therefore, the results reported by Guo et al. (2009) exhibit a similar trend and mostly fall between the data represented by the rectangles with dashed lines and the circles with solid lines. For the experimental data of Koushki et al. (2023), only the partially submerged cases are included due to its high value of the obstacle height (h). Overall, the present numerical results agreed well with the experimental data and the exising literature data.

Fig. 8(a) presents the shear stress distribution under different obstacle elevations at the initial stage before significant scour develops. The shaded area indicates the zone beneath the obstacle. As shown in Fig. 8(a), the shear stress for $e/y = 0.05$ is lower than that for $e/y = 0.15$, and it decreases further with increasing of e/y. For example, the cases with $e/y = 0.15$, 0.325, 0.6 and 0.825, show a continuous decline in shear stress. This trend is closely related to the relationship between

Table 1
Test conditions with varying obstacle elevations.

Solid obstacle										
Case No.	0	1	2	3	4	5	6	7	8	9
e/y	0.05	0.15	0.225	0.325	0.425	0.475	0.6	0.725	0.825	0.9
constant $h/y = 0.275$, and $l/y = 1.15$										

Table 2
Test conditions with varying obstacle heights and lengths.

Solid obstacle										
Case No.	0a	1a	3a	7a	1b	3b	6b	0c	3c	6c
e/y	0.05	0.15	0.325	0.725	0.15	0.325	0.6	0.05	0.325	0.6
h/y	0.15 (reduced)				0.35 (increased)			0.275		
l/y	1.15							0.85 (reduced)		

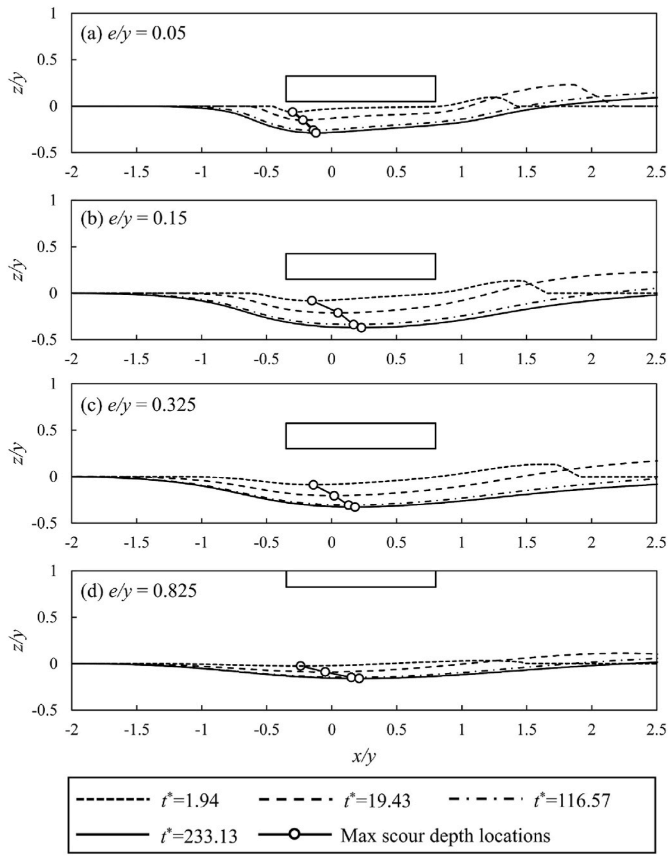

Fig. 5. Evolution of scour profiles and maximum scour locations with constant values of $h/y = 0.275$, $l/y = 1.15$ and varying obstacle elevations: (a) $e/y = 0.05$; (b) $e/y = 0.15$; (c) $e/y = 0.325$; (d) $e/y = 0.825$.

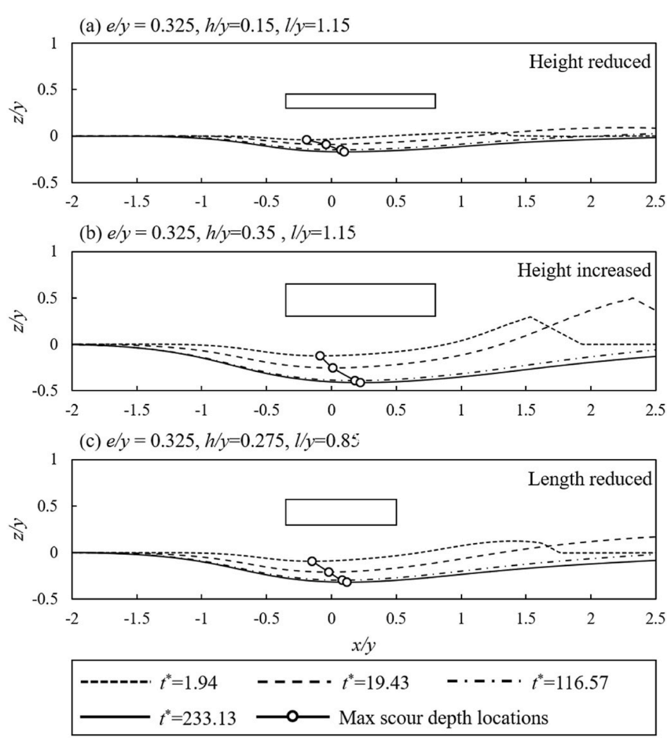

Fig. 6. Evolution of scour profiles and maximum scour locations: (a) obstacle height reduced ($h/y = 0.15$); (b) obstacle height increased ($h/y = 0.35$); (c) Obstacle length reduced ($l/y = 0.725$).

scour depth and obstacle elevation and explains why the maximum scour depth occurs at $e/y = 0.15$.

Fig. 8(b)–(d) show snapshots of the flow distribution at an early stage of scour ($t^* = 19.43$) under obstacle elevations of $e/y = 0.15$, 0.05, and 0.03, respectively. In Fig. 8(b), when $e/y = 0.15$, the channel beneath the obstacle is sufficiently high for both flow and scoured sediment to pass through. In Fig. 8(c), at $e/y = 0.05$, the reduced channel height, together with a downstream dune formed by deposited sediment near the outlet gap, partially blocks the flow passage and limits further scour. When the obstacle elevation is extremely small, as shown in Fig. 8(d), the low-velocity wake region lies close to the bed, promoting the growth of a larger downstream dune. As scour progresses, this dune may further block the outlet gap, preventing flow from passing through. This blockage effect by the downstream dune further explains why in the low obstacle elevation regime, smaller e/y values correspond to smaller shear stress values, and thereby smaller scour depths.

Fig. 9 shows snapshots of the velocity distribution around a submerged solid obstacle, where blue represents low velocity and orange denotes high velocity. As shown in Fig. 9(a), due to flow separation near the obstacle, a low-velocity recirculation zone develops immediately beneath the solid obstacle. To satisfy mass continuity, the flow below the obstacle redistributes vertically, where the upper part of the profile is slowed by separation, the lower part accelerates, forming a high-velocity zone close to the bed. This redistribution of velocity leads to a nonuniform bed shear stress distribution along the channel beneath the obstacle, as shown in Fig. 8(a). For instance, in regions affected by flow separation, the locally elevated near-bed velocity contributes to a peak of the bed shear stress shown in Fig. 9(a), which in turn intensify sediment transport and erosion. At the initial stage, the high velocity zone is located near the upstream edge of the obstacle, where scour is first initiated. Once a scour hole develops, as shown in Fig. 9(b), the average velocity and bed shear stress at the maximum scour location are relieved due to the increased vertical distance. Meanwhile, the shear stress beneath the obstacle intensifies in the downstream direction. Consequently, the zone of higher velocity and shear stress shifts slightly downstream, leading to a downstream movement of the maximum scour location. By the end of the simulation, the maximum scour depth is positioned near the obstacle centerline, and its downstream migration has nearly ceased.

5. Numerical simulations of scour beneath porous obstacles

Pressure scour beneath porous obstacles has not been well addressed in the existing studies. Thus, numerical simulations for scour beneath porous obstacles are investigated in this section. Varying obstacle porosities (φ) and elevations (e/y) are studied. As listed in Table 3, a total of 27 cases are investigated in this research, including three different porosities, $\varphi = 25$ %, $\varphi = 40$ % and $\varphi = 50$ %, with varying e/y values ranging from 0.05 to 0.9. Obstacle porosity of $\varphi = 75$ % has also been investigated, but insignificant scour is observed in the present study.

5.1. *Scour profiles and locations of maximum scour depth beneath porous obstacles*

Fig. 10 presents typical scour profiles in four sub-figures corresponding to different initial elevations e/y for porous obstacles. In each sub-figure, scour profiles for three different porosities are illustrated. The solid, dashed and dotted lines represent the scour profile of porosities of $\varphi = 25$ %, 40 %, and 50 %, respectively. The black line with circular symbols indicates the location of the maximum scour depth. Additionally, the blue line with rectangular symbol represents the scour profile and maximum scour location of solid obstacle at equilibrium stage. In Fig. 10(a)–(d), it is evident that the maximum scour for solid obstacles are all located near the centerline, while for porous obstacles, the maximum scour depths are shifting to the downstream edge, such differences result in a steeper upstream scour hole slope beneath the

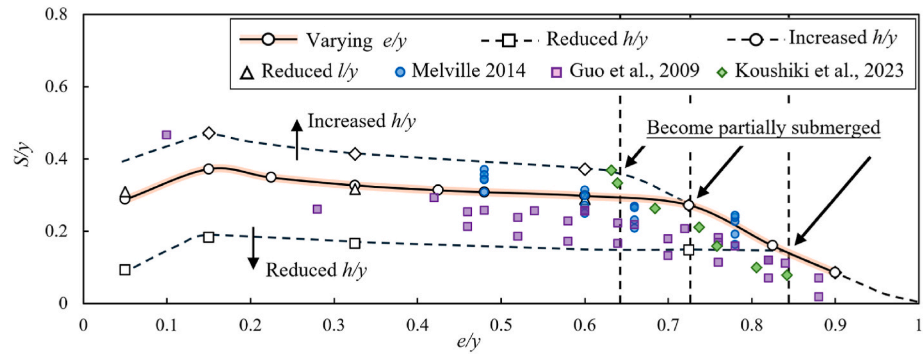

Fig. 7. Maximum scour depth under varying obstacle elevations and dimensions compared with data from the existing literature.

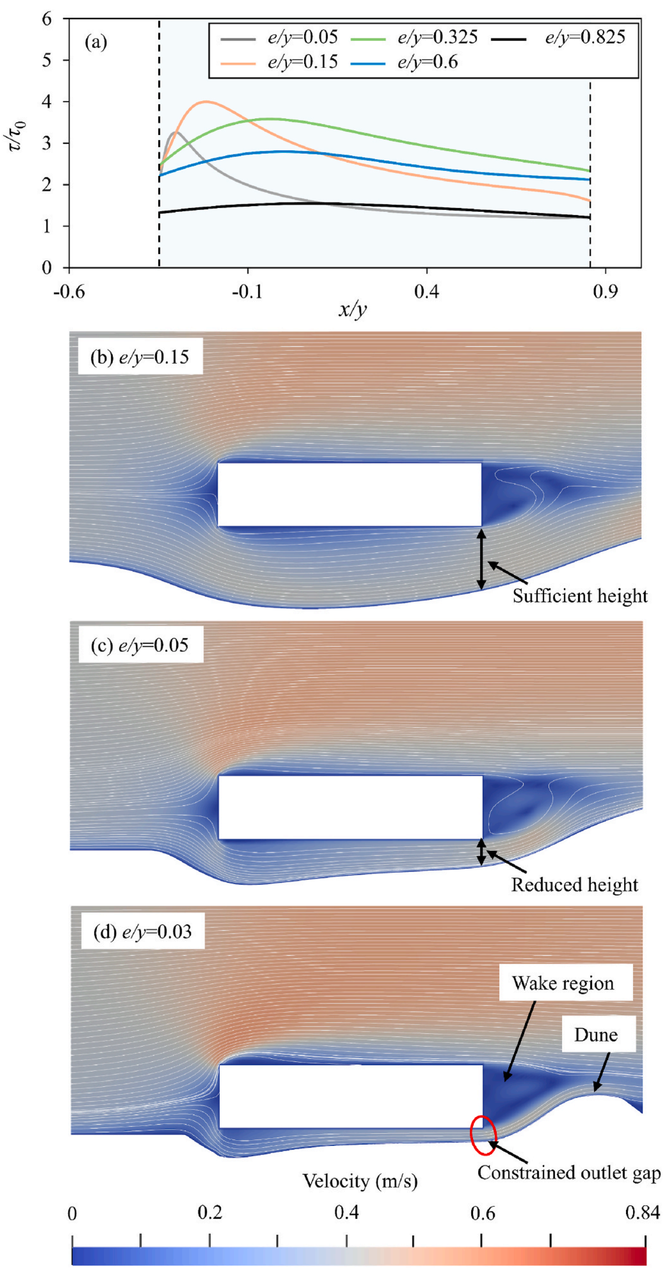

Fig. 8. (a) Initial shear stress distribution under different obstacle elevations; and snapshots of flow distribution at early scour stage ($t^* = 19.43$) for obstacle elevations of (b) $e/y = 0.15$, (c) $e/y = 0.05$, and (d) $e/y = 0.03$.

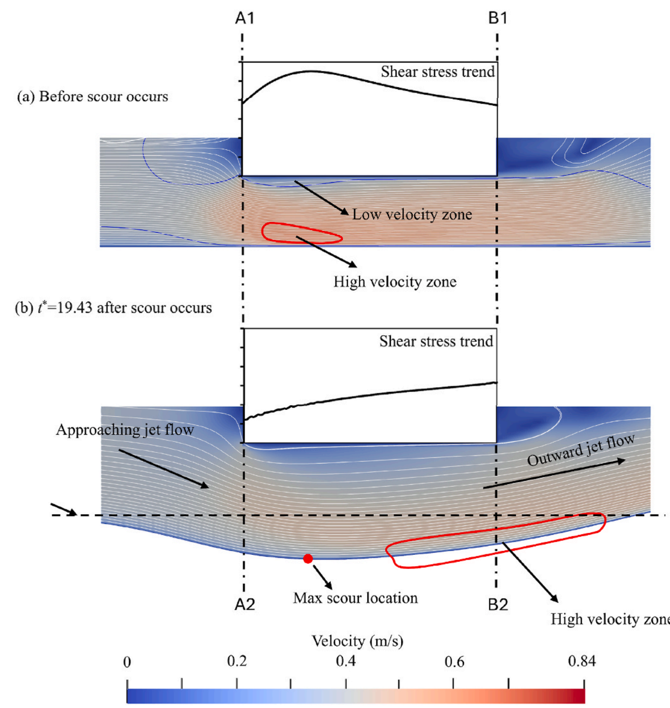

Fig. 9. Flow distribution under $e/y = 0.325$: (a) before scour occurs; (b) after scour occurs ($t^* = 19.43$).

solid obstacles than the porous ones. With increasing porosity, the scour profiles become less pronounced, and the slope of the upstream scour hole is reduced accordingly. Furthermore, compared to the scour profiles of the solid obstacles, the downstream side scour hole slopes of porous obstacles are flatter as well. Overall, the scour profiles of porous obstacles are generally gentler and less significant compared to solid obstacles. This occurs because higher porosity allows a greater proportion of flow to pass through the obstacle, thereby reducing the flow rate beneath it. As a result, the average velocity and bed shear stress are both decreased.

Additionally, unlike pier scour, where the maximum depth typically occurs adjacent to the pier surface, the location of maximum scour in vertical contraction scour is influenced by obstacle porosity, and as discussed earlier, shifts across different scour stages. Accordingly, reinforcement may need to account for variations in location.

5.2. Maximum scour depth and velocity distribution of porous obstacles

As shown in Fig. 11(a), the orange curve denotes the scour depth of solid obstacles at various e/y values. While the rectangular, circular and triangular symbols represent the porosity of $\varphi = 25$ %, $\varphi = 40$ % and $\varphi =$

Table 3
Test conditions of scour beneath porous obstacles.

Porous obstacle ($h/y = 0.275$, and $l/y = 1.15$)									
Case No. ($\varphi = 25$ %)	10	11	12	13	14	15	16	17	18
e/y	0.09	0.15	0.225	0.325	0.425	0.475	0.6	0.725	0.9
Case No. ($\varphi = 40$ %)	19	20	21	22	23	24	25	26	27
e/y	0.05	0.15	0.225	0.325	0.425	0.475	0.6	0.725	0.9
Case No. ($\varphi = 50$ %)	28	29	30	31	32	33	34	35	36
e/y	0.05	0.15	0.225	0.325	0.425	0.475	0.6	0.725	0.9

*Cases with $\varphi = 75$ % were also examined. However, no significant scour depths were detected.

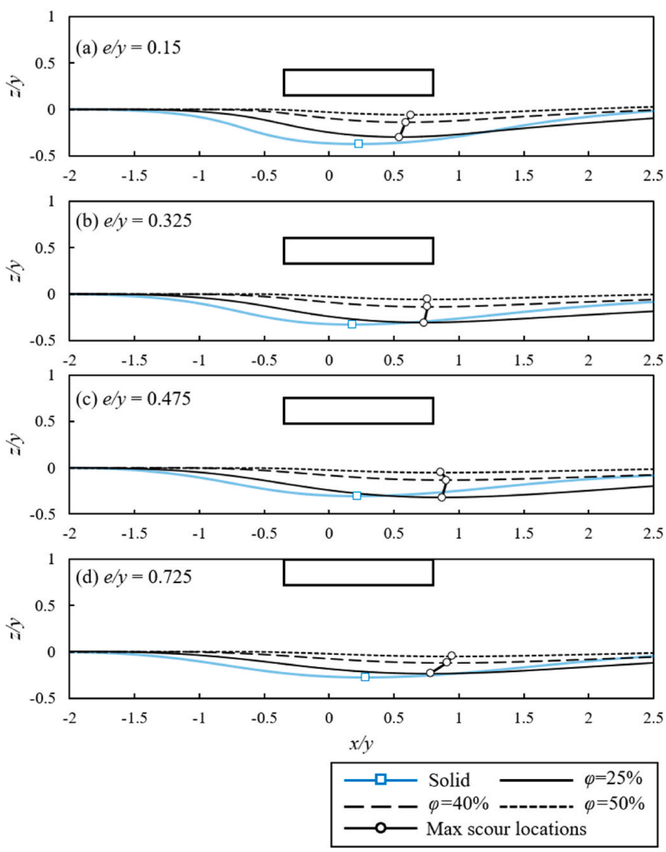

Fig. 10. Scour profiles at equilibrium stage for different porosities under various e/y values: (a) $e/y = 0.15$; (b) $e/y = 0.325$; (c) $e/y = 0.475$; and (d) $e/y = 0.725$.

50 %, respectively. The dashed line indicates the point after which the obstacle becomes partially submerged. When the porous obstacle is fully submerged, the scour depth is less sensitive to the variations in the initial gap ratio e/y compared to solid obstacles. As shown in Fig. 11(a), for each porosity, scour depths show insignificant variation within the range $0.15 < e/y < 0.6$. In the low e/y region, e.g., $e/y = 0.05$, the scour depths are slightly smaller than those observed in other fully submerged scenarios, consistent with the trend observed for solid obstacles. However, once $e/y > 0.725$, the porous obstacle becomes partially submerged, and scour depths begin to decrease rapidly with further increases in e/y.

As shown in Fig. 11(b), the solid obstacle is considered as a special case with porosity $\varphi = 0$. It is evident that an increasing porosity leads to a reduction in scour depth. When $\varphi = 25$ %, the scour depth is nearly equivalent to that of the solid obstacle, suggesting that low-porosity obstacles offer limited reduction in scour depth compared to solid ones. However, a significant decrease in scour depth is observed as porosity increases from $\varphi = 25$ %–40 %, and from $\varphi = 40$ %–50 %. Furthermore, simulations conducted with $\varphi = 75$ % show insignificant scour depths, indicating that vertical contraction scour is unlikely to occur beneath high porous obstacles.

The effects of e/y and φ on the maximum bed shear stress (τ_m) are presented in Fig. 11(a) and Fig. 11(b), respectively. The maximum bed shear stress is obtained before the morphology model is activated. As shown in the figure, the variation in maximum bed shear stress follows nearly the same trend as the scour depth. This indicates that the maximum bed shear stress can serve as a useful indicator for estimating the potential scour intensity for different obstacle configurations.

Fig. 12(a)–(d) shows the velocity fields for porous obstacles with $\varphi = 40$ % at initial gap ratios $e/y = 0.05$, 0.225, 0.6, and 0.725, respectively. Five velocity profiles are illustrated, located at x/y = -0.35, −0.06, 0.23, 0.51, and 0.8. As shown in Fig. 12(a), when e/y is very small, the average velocity beneath the obstacle is restricted, similar to the solid obstacle case. This results in lower shear stress and a smaller scour depth at $e/y = 0.05$ compared with $e/y = 0.15$. A pronounced velocity reduction is also evident within the porous obstacle itself. Because a portion of the flow penetrates the porous obstacle, the average velocities both beneath and above it are lower than those for solid obstacles, leading to overall smaller scour depths. Only minor velocity changes occur between the initial and equilibrium stages. Furthermore, across Fig. 12(a)–(d), the velocity within the porous medium decreases along with the flow direction, producing progressively higher velocities beneath the downstream of the obstacle. This pattern explains why the maximum scour consistently develops near the downstream edge of the porous obstacle.

Fig. 13 illustrates the influence of porosity on the initial velocity distribution. The green, black, blue, and red curves represent the velocity profiles for solid, $\varphi = 25$ %, $\varphi = 40$ %, and $\varphi = 50$ %, respectively. As shown in Fig. 13(a) and (b), lower porosity results in a greater reduction in flow velocity. Specifically, as porosity increases, the velocities at both the top and bottom edges of the porous obstacle increase, indicating reduced flow resistance. Additionally, since no flow can pass through solid obstacles, the flow velocity beneath and above the solid obstacle is higher than that around porous obstacle.

6. Theoretical analysis of scour beneath solid and porous obstacles

6.1. Theoretical analysis of initial flow rate distribution of solid obstacles

In Fig. 7, a turning point at the low elevation condition is observed approximately at $e/y = 0.15$, where the highest scour depths can be detected. This observation suggests that scour development may be constrained under a lower elevation condition. The presence of the obstacle may alter the flow rate and velocity distribution above and beneath it, thereby modifying the shear stress acting on the sand bed and resulting in variations in scour depth. Furthermore, variations in obstacle height and length may also modify the flow rate distribution and bed shear stress, thereby influencing the resulting scour depth. Overall, the flow rate beneath the obstacle tends to be the dominant factor. The following section will discuss this theoretically.

As shown in Fig. 14, Q denotes the total discharge, where u_1 and q_1 represent the flow velocity and discharge above the obstacle, u_2 and q_2 represent those below the obstacle. In the present two-dimensional cases, the flow above the obstacle is treated as open channel flow,

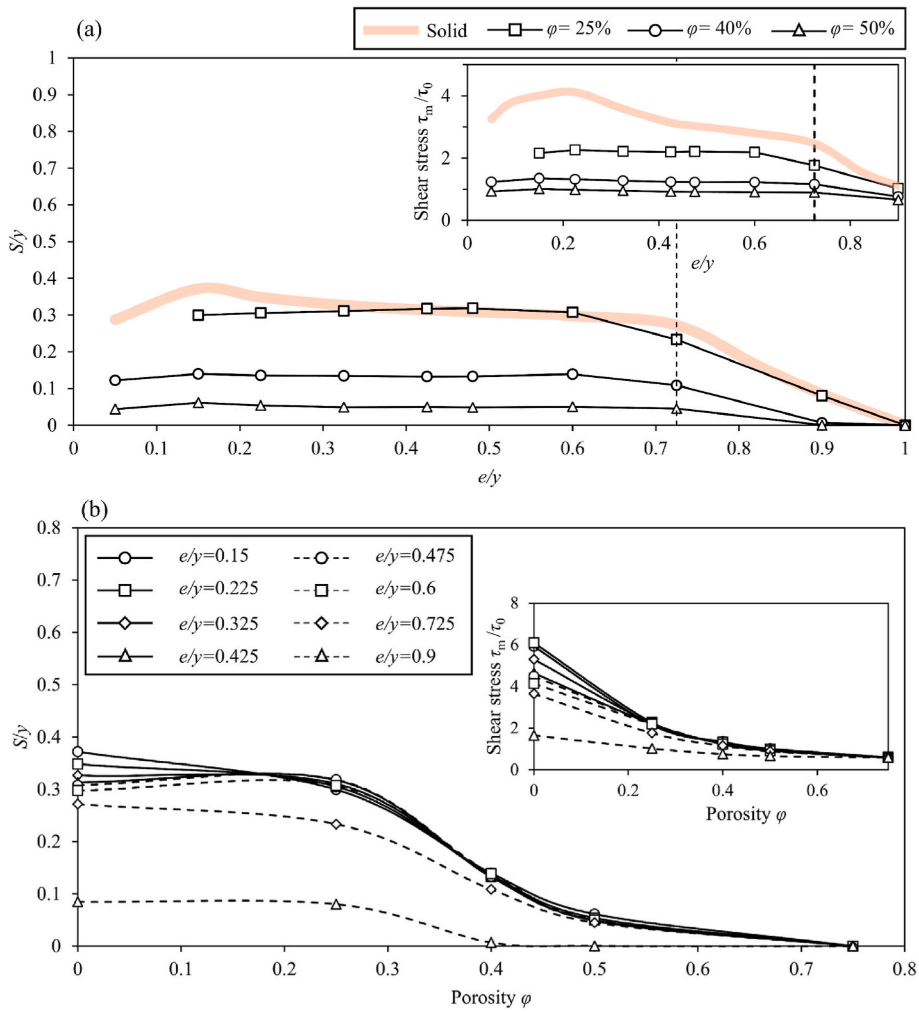

Fig. 11. Maximum scour depths of porous obstacles: (a) with varying e/y; (b) with different porosities.

while the flow beneath the obstacle is considered as pipe flow. Accordingly, the theoretical analysis assumes that the approaching flow is steady and uniform, and that it is vertically irrotational, with viscous effects represented through flow resistance. From section I-I to section III-III, the head loss above and beneath the obstacle should be the same.

Therefore, using the classic Darcy-Weisbach equation (neglecting the minor head loss), we can get:

$$f_1 \frac{l}{4R_1} \frac{u_1^2}{2g} = f_2 \frac{l}{4R_2} \frac{u_2^2}{2g} \tag{32}$$

where f_1 and f_2 are the friction head loss coefficients; R_1 and R_2 are the hydraulic radius. The following relationship can be easily derived by Eq. (32):

$$\frac{u_1}{u_2} = \sqrt{\frac{R_1}{R_2} \frac{f_2}{f_1}} \tag{33}$$

As mentioned before, in the present two-dimensional cases, the flow above the obstacle is assumed as open channel flow while the flow beneath the obstacle is considered as pipe flow. Thus, $R_1 = w$, and $R_2 = e/2$. Moreover, q_1, q_2 can be calculated by $v_1 w$ and $v_2 e$, respectively. Therefore, the following relationship can be obtained:

$$\frac{q_1}{q_2} = \left(\frac{w}{e}\right) \sqrt{\frac{R_1}{R_2} \frac{f_2}{f_1}} = \frac{\sqrt{2}}{2} \left(\frac{w}{e}\right)^{3/2} \sqrt{\frac{f_2}{f_1}} \tag{34}$$

the friction head loss coefficient f can also be expressed using the Manning n:

$$f = \frac{8gn^2}{R^{1/3}} \tag{35}$$

by substituting Eq. (35) into Eq. (34), the following relationship can be derived:

$$\frac{q_1}{q_2} = \left(\frac{w}{e}\right) \sqrt{\frac{R_1}{R_2} \frac{f_2}{f_1}} = \frac{\sqrt{2}}{2} \left(\frac{w}{e}\right)^{5/3} \frac{n_2}{n_1} \tag{36}$$

to estimate Manning coefficient n, multiple empirical equations were proposed. Some of the well-known methods are summarized and listed in the following table (Table 4).

For simplicity and applicability, Manning coefficient n is treated as a function of $d^{1/6}$ in the present study, e.g., using estimation method by Julien (2010).

The solid obstacle surface is considered as smooth wall, and its corresponding grain size d_1 is 0.01 mm. However, the channel beneath the solid obstacle has a smooth top and a rough bottom. Thus, an averaged value is utilized as the equivalent grain size $d_2' = 0.425$ mm. Therefore, we can get:

$$\frac{n_2}{n_1} = \left(\frac{d_2'}{d_1}\right)^{1/6} \approx 1.8 \tag{37}$$

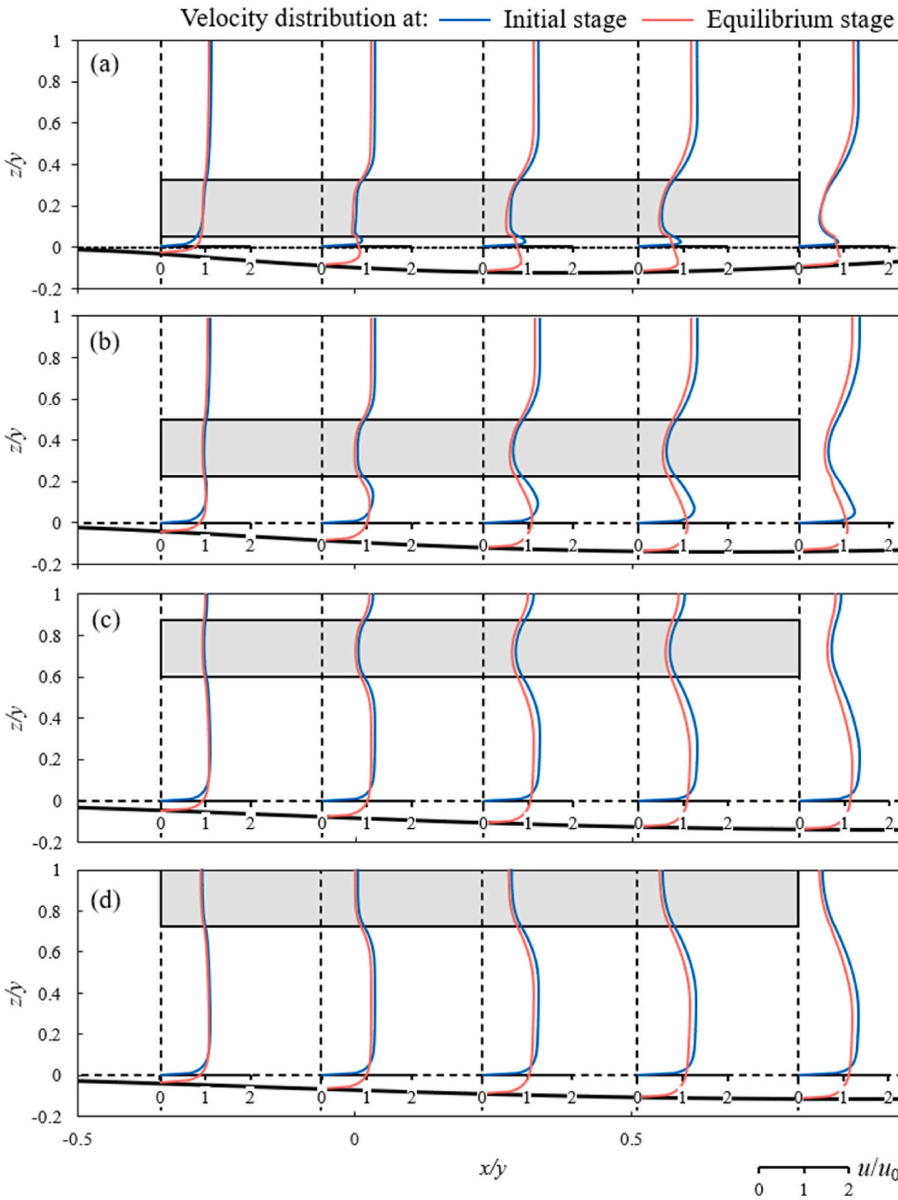

Fig. 12. Velocity distribution of porous obstacles (φ = 40 %): (a) e/y = 0.05; (b) e/y = 0.225; (c) e/y = 0.6; (d) e/y = 0.725.

$$\frac{q_1}{q_2} = 1.8 \times \frac{\sqrt{2}}{2}\left(\frac{w}{e}\right)^{5/3} = 1.27\left(\frac{w}{e}\right)^{5/3} \tag{38}$$

$$\frac{q_2}{Q} = \frac{q_2}{q_1 + q_2} = \frac{1}{1 + 1.27\left(\frac{w}{e}\right)^{5/3}} = \frac{1}{1 + 1.27\left(\frac{e}{w}\right)^{-5/3}} \tag{39}$$

The theoretical results calculated by Eq. (39) is plotted against the numerical results in Fig. 15. The dashed curve represents the original theoretical results, and the circles, rectangles, and triangles indicate the numerical results obtained from the present study under various e/y, h/y, and l/y conditions, respectively. A good agreement is observed between the theoretical predictions and the numerical data. As indicated by the circular markers, a decrease in obstacle elevation e leads to a corresponding decrease in e/w, resulting in a rapid decline in the ratio of q_2/Q. When obstacle elevation e becomes sufficiently small, q_2 only accounts for a minimal portion of the total flow rate. This result confirms that limited obstacle elevation e/y values with roughness bottom can restrict the flow through the narrow gap, subsequently affecting the velocity field and shear stress, and ultimately influencing vertical contraction scour. Additional simulations with varying obstacle height (h/y) and length (l/y) are conducted and compared with the theoretical results, as indicated by the rectangles and triangles in Fig. 15. Given the relationship $y = w + h + e$, where the flow depth y is constant, changes in obstacle height (h/y) result in variations in the ratio e/w, which in turn affect the flow rate distribution beneath the solid obstacle. However, changes in obstacle length (l/y) do not alter the e/w ratio and have only a limited influence on the flow rate distribution.

Eq. (40) demonstrates a basic framework of the initial flow rate distribution, which is:

$$\frac{q_2}{Q} = \frac{1}{1 + a\left(\frac{e}{w}\right)^{-b}} \tag{40}$$

The coefficients a and b are based on the previously established empirical formulas and assumptions. For example, coefficient a is strongly dependent on Manning n, which is calculated by empirical formulas. Additionally, coefficient b is related to the $R^{-1/3}$ term of Eq. (35), the exponent (i.e., $-1/3$) comes from the classic Manning equation,

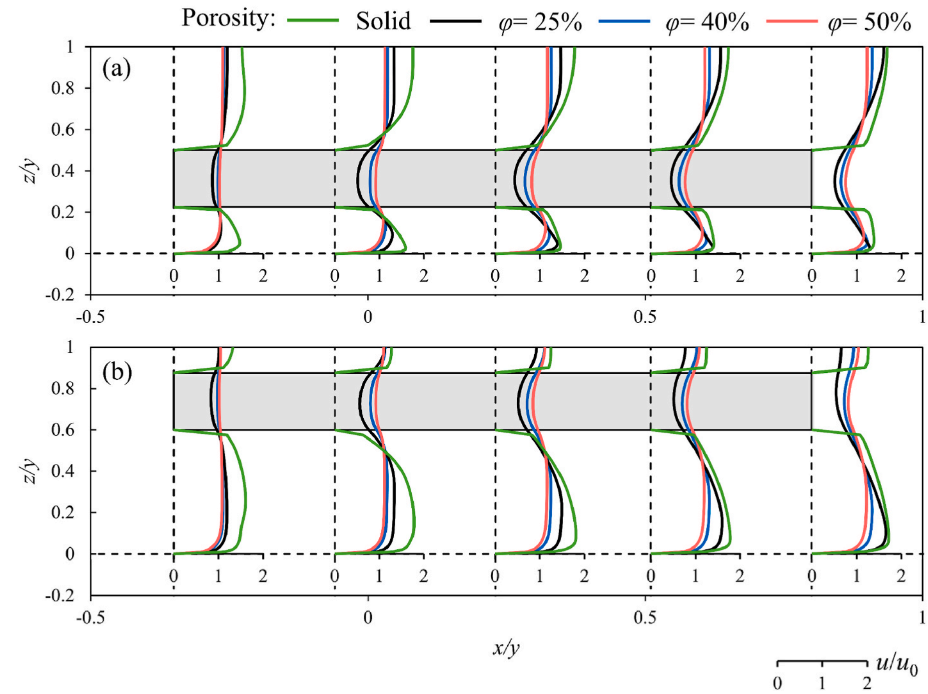

Fig. 13. Velocity distribution of obstacles with different porosities under different obstacle elevations: (a) $e/y = 0.225$; (b) $e/y = 0.6$.

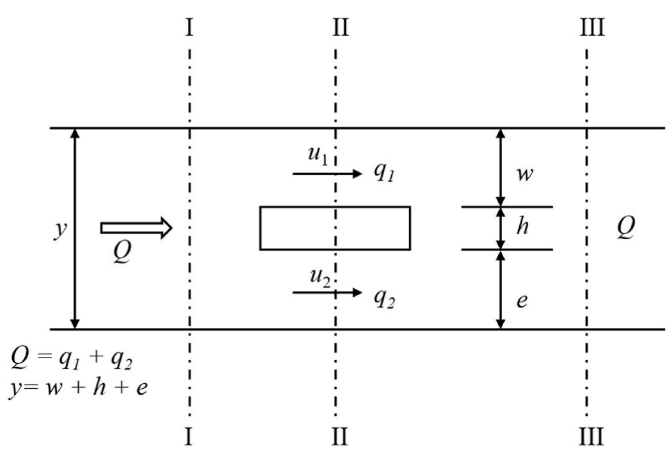

Fig. 14. Schematic drawing of the initial flow rate distribution.

Table 4
A summary of estimation methods for Manning coefficient n.

$n = \frac{1}{21.1} d^{1/6} = 0.047 d^{1/6}$ *or* $n = 0.047 (k_N)^{1/6}$	Strickler (1923)
$n = 0.062 d^{1/6}$ when $y/d > 100$	Julien (2010)
$n = 0.39 S^{0.38} R^{-0.16}$ when in steep channel	Jarrett (1984)
$n = \frac{0.0926 R^{1/6}}{1.16 + 2.0 \log\left(\frac{R}{d_{84}}\right)}$ for gravel/boulder streams	Limerinos (1970)
$n = \frac{R^{1/6}}{\sqrt{g} 3.54 \left(\frac{y}{d}\right)^{0.287}}$ for gravel bed	Griffiths (1981)

$$u = \frac{1}{n} R^{2/3} S^{1/2} \tag{41}$$

where the exponent (i.e., 2/3) is also empirical. Moreover, assumptions and simplifications are also included in the analysis. Therefore, slight adjustments to these coefficients may better capture the relationship between q_2/Q and e/w in the present scenarios. Eventually, modified coefficient values of $a = 1.1$ and $b = 1.2$ are adopted in further analysis. The modified curve with updated coefficients is shown as the solid line in Fig. 15 and demonstrates improved agreement with the numerical results.

As shown in Fig. 16(a), the average flow velocity at different obstacle elevations are plotted. Additional simulations, same as those presented in Fig. 15, are conducted by varying the obstacle height (h/y) and length (l/y). As indicated by the rectangles in Fig. 16(a), varying obstacle height (h/y) changes the average velocity beneath the solid obstacle and ultimately influence the scour depth. In contrast, as shown by the triangles in Fig. 16(a), variations in obstacle length (l/y) do not change the e/w ratio and have a limited effect on the flow rate and average flow velocity beneath the obstacle. Therefore, no significant influence on scour depth should be observed by varying the obstacle length.

The friction velocity u_* and shear stress on the bottom are the variables which can directly influence the local scour depth. For uniform flow, we can obtain:

$$u_* = u_2 \sqrt{\frac{f_2}{8}} = \frac{u_2 n_2}{\left(\frac{e}{2}\right)^{1/6}} \sqrt{g} = \alpha_2 \frac{u_2 n_2}{e^{1/6}} \tag{42}$$

Manning coefficient n can be treated as a function of $d^{1/6}$ and d is constant in the present study. When combining the effect of rough bed and smooth wall, the equivalent grain size is 0.425 mm for solid obstacles. Thus, Eq. (42) can be written as:

$$u_* = 0.189 \frac{u_2}{e^{1/6}} \tag{43}$$

$$\tau = \rho u_*^{\,2} \tag{44}$$

The shear stress (τ) is illustrated in Fig. 16(b). The shear stress of the theoretical results agrees well with the average shear stress from the

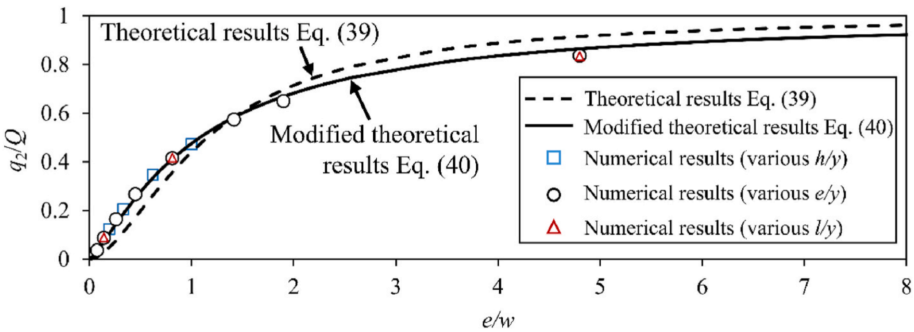

Fig. 15. Theoretical and numerical flow rate distribution at initial stage with various e/w values.

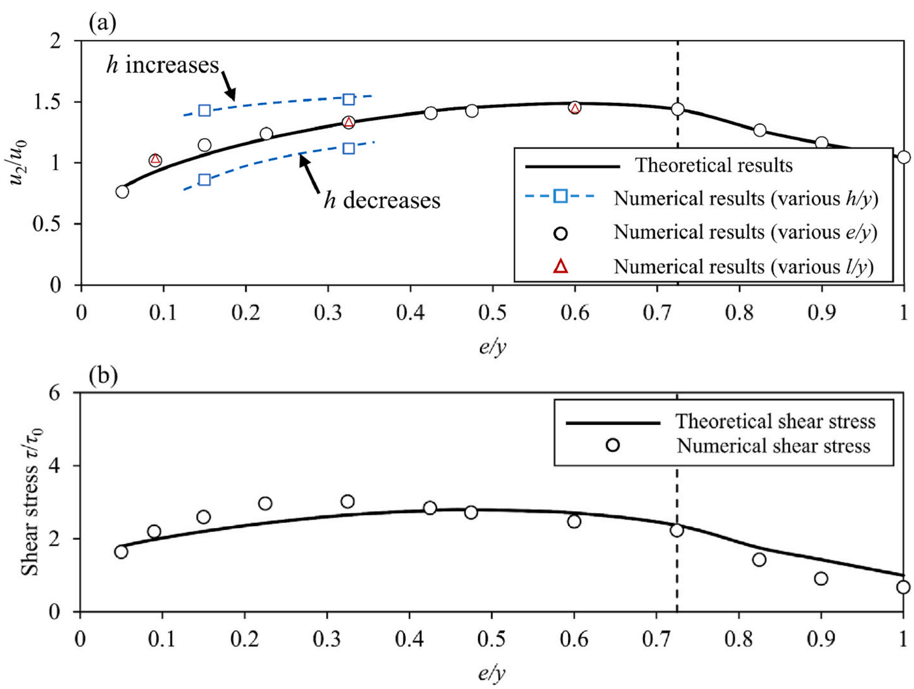

Fig. 16. Theoretical analysis of solid obstacles: (a) theoretical and numerical average flow velocity beneath the solid obstacle at initial stage; (b) theoretical shear stress trend at initial stage.

numerical results, and also algin with the trend in scour depth variation in Fig. 7. For example, an increase and then decrease trend with increasing e/y, and a rapid decline in shear stress occurs when $e/y > 0.725$, aligning with the decreasing trend in scour depths within the same range shown in Fig. 7.

As mentioned before, theoretical analysis is based on several simplifying assumptions. For example, Eq. (42) is typically derived for uniform flow conditions, whereas the flow beneath the solid obstacle is non-uniform due to flow separation, particularly on the upstream side. The average flow velocity obtained from the simulation shows excellent agreement with the theoretical analysis, whereas the shear stress exhibits a slight peak shift. This peak shift is likely attributed to flow separation, which is not considered in the theoretical analysis. Scour depth is governed by complex, dynamic interactions influenced by factors such as initial flow distribution, flow separation, and morphological evolution. While the present theoretical analysis effectively captures the flow rate distribution and average velocity beneath the obstacle, well presents the turning point and the reduction trend in scour depth at low e/y value conditions as well as the rapid decline when the obstacle becomes partially submerged, e.g., $e/y > 0.725$.

6.2. Theoretical analysis of porous obstacles

Though the surface roughness of porous obstacles is hard to obtain, some derived relationships, such as Eq. (40) can be used for porous obstacles with further modifications. Since the flow distribution along the porous obstacle may vary slightly, the profile at $x/y = 0.23$, near the center of the obstacle, is selected as a representative for the following analysis. Eq. (40) gives out a basic relationship form between q_2/Q and e/w, for porous obstacles, a modification coefficient α_φ is applied, so Eq. (41) can be written as:

$$\frac{q_2}{Q} = \frac{\alpha_\varphi}{1 + a\left(\frac{e}{w}\right)^{-b}} \tag{45}$$

where α_φ can be derived from the data obtained in the present simulations. Fig. 17 (a) shows the flow rate distribution of porous obstacles at the initial stage. The α_φ presented in Fig. 17(a) is based on the fitting results of Fig. 17(b), (c) and (d). Additionally, in order to achieve a better fitting result, coefficient b is modified from 1.2 to 1.1 for porous obstacles. As shown in Fig. 17(a), when the porosity φ equals to 25 %, 40 %, and 50 %, α_φ is 0.86, 0.8, and 0.76, respectively. Therefore, a simple linear relationship can be derived between α_φ and φ, which is:

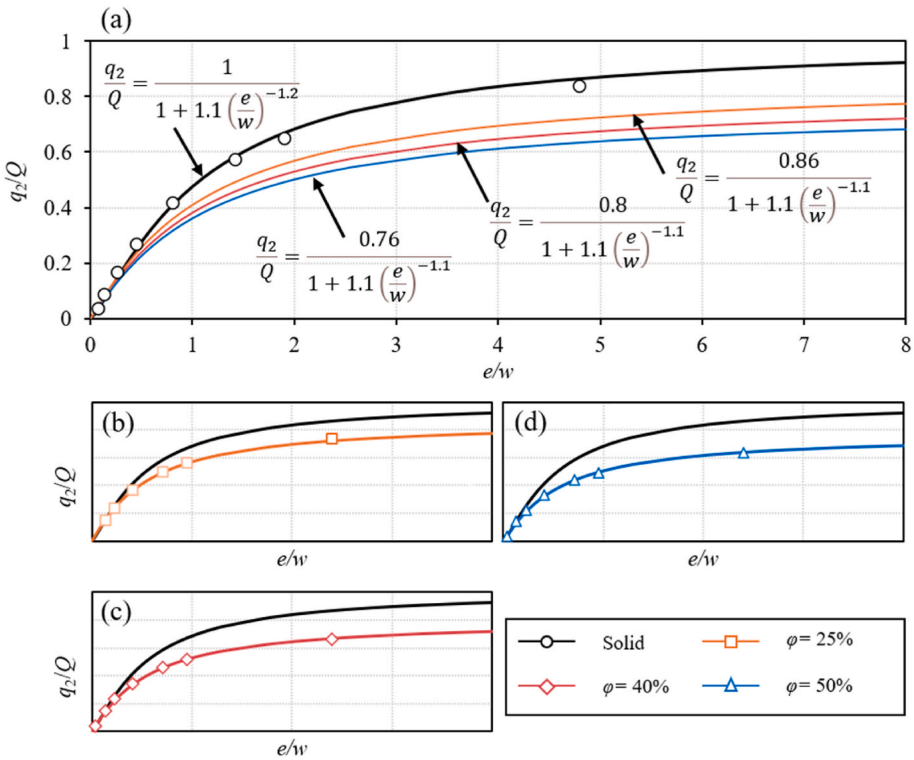

Fig. 17. Flow rate distribution for porous obstacles (at $x/y = 0.23$): (a) comparison with solid obstacles; (b) fitting of $\varphi = 25$ %; (c) fitting of $\varphi = 40$ %; (d) fitting of $\varphi = 50$ %.

$$\alpha_\varphi = 0.96 - 0.4\varphi \tag{46}$$

so, Eq. (46) can be written as follows:

$$\frac{q_2}{Q} = \frac{0.96 - 0.4\varphi}{1 + 1.1\left(\frac{e}{w}\right)^{-1.1}} \tag{47}$$

Eq. (47) can be used for estimating initial flow rate distribution of porous obstacles. Since the range of porosity of the present study is 25 % $< \varphi <$ 50 %, Eq. (47) is expected to show best performance within this range. For porous obstacles out of this range, the performance of Eq. (47) requires validation by further studies.

Fig. 18 compares the average flow velocity and shear stress beneath the porous obstacles between the theoretical predictions and numerical results under the fully submerged conditions. As shown in Fig. 18(a), the simulated flow velocities show good agreement with the theoretical solutions. In Fig. 18(b), specific assumptions are applied to estimate the bed shear stress beneath the porous obstacle. The overlapping blue and black solid curves indicate that both open-channel flow and original pipe-flow assumptions predict a significant amplification of shear stress. As previously shown in Figs. 12 and 13, the velocity at the bottom of the porous obstacle is lower than the free-surface (or slip wall) velocity but remains clearly above zero, unlike the condition at a no-slip wall boundary, and lower porosity induced higher velocity reduction. To account for this effect, an effective virtual roughness height ($k_v < 0$) is introduced. Although this value does not represent a physical roughness, it serves to adjust the velocity profile shape in the analytical pipe-flow model, thereby providing a better approximation of shear stress behavior over a porous surface.

By introducing k_v, the corresponding virtual equivalent grain size d_v (<0) can be obtained. This value is then used to calculate an equivalent Manning n for the analytical pipe-flow model using the method by Julien (2010), consistent with the approach previously applied to the solid obstacle cases. Subsequently, the friction velocity and shear stress can be estimated using the same approach, as described in Eqs. 42–44. As shown by the dashed black lines in Fig. 18, the theoretical results of shear stress align good with the numerical data when the equivalent grain sizes d_v are set to -0.24 mm, -0.64 mm, and -0.72 mm for porous obstacles with $\varphi = 25$ %, $\varphi = 40$ % and $\varphi = 50$ %, respectively. Notet that when d_v are set to -0.24 mm, -0.64 mm, and -0.72 mm, the equivalent grain sizes for the analytical pipe-flow model are 0.3 mm, 0.1 mm, and 0.06 mm, respectively. Additionally, both the theoretical and numerical results for shear stress in Fig. 18(b) show minimal variation across different initial elevations (e/y) under fully submerged conditions. This observation aligns with the simulated scour depths, which also show negligible changes with varying e/y values in the fully submerged regime. As discussed earlier in Fig. 11, the maximum bed shear stress is the main indicator for sediment transport intensity and scour depth. Although the present theoretical model does not account for flow separation and therefore predicts only the average bed shear stress, the numerical results in Fig. 18(b) indicate that the average bed shear stress shown by the black circles and the maximum bed shear stress shown by the purple rectangles develop in almost the same manner when parameters such as e/y and φ change. This means that the variation of the theoretical average bed shear stress can also serve as a useful indicator for the general trend of scour depth.

Overall, the theoretical analysis of flow rate distribution, average velocity, and average bed shear stress beneath solid and porous obstacles provides a quick reference for the general trends in scour depth variation. However, accurate prediction of local scour requires high-resolution CFD simulations.

7. Discussions

Based on the numerical results and theoretical analysis, a general trend can be derived to represent the fundamental relationship between the obstacle elevation (e/y) and the maximum scour depth (S/y). As shown in Fig. 19, the black curve indicates the solid obstacle, and the blue one denotes the porous obstacle. For the solid obstacles, it is evident that no scour can be detected when the obstacle is completely out of the

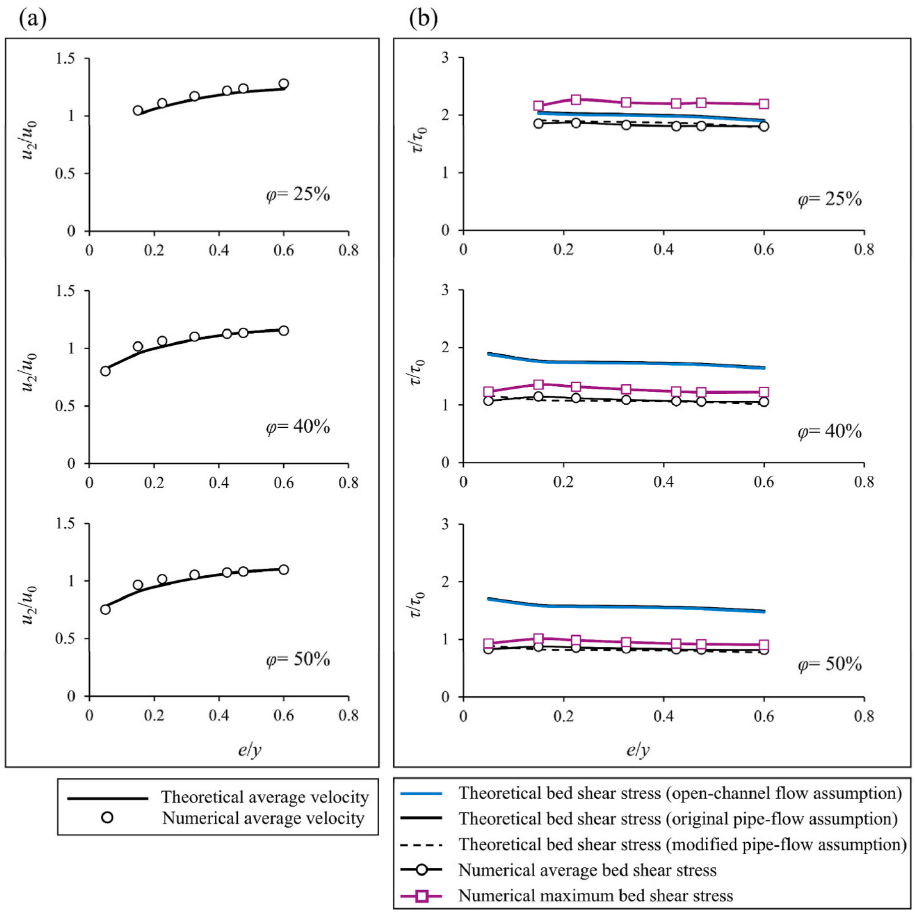

Fig. 18. Theoretical analysis of porous obstacles: (a) theoretical and numerical results of average flow velocity at initial stage; (b) theoretical and numerical results of shear stress at initial stage.

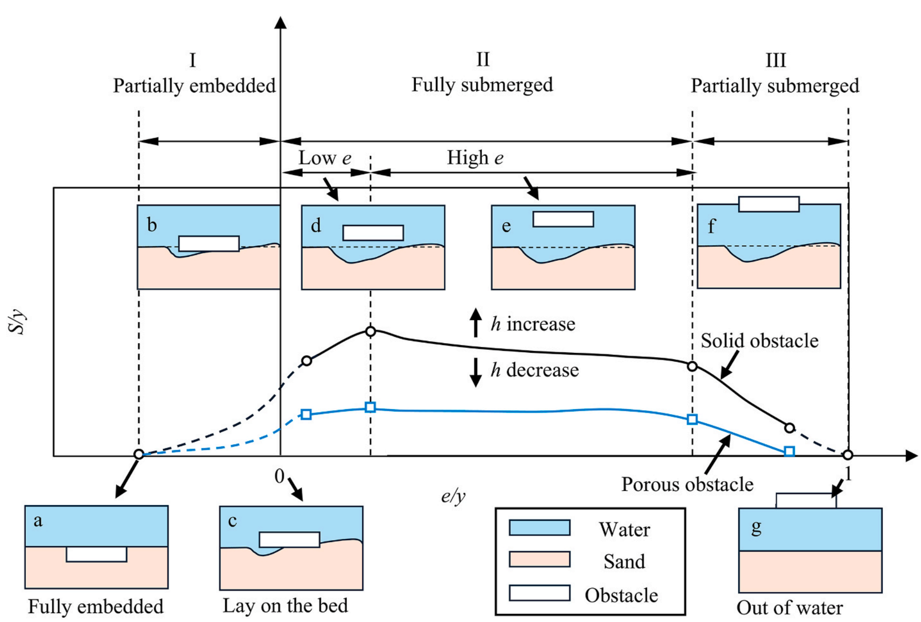

Fig. 19. Basic relationships between initial elevation and maximum scour depth for both solid and porous obstacles.

flow at both ends, e.g., when it is fully embedded in the sand bed (type-a) or entirely above the water surface (type-g). Therefore, at $e/h = -1$ (corresponding $e/y = -0.275$ in the present case) and $e/y = 1$, the values of S/y are zero in Fig. 19. From the partially embedded condition (type-b) to the case where the obstacle lies directly on the bed (type-c), although the initial shear stress beneath the obstacle is zero, edge scour may occur on the upstream side and undercutting may gradually develop beneath the obstacle. This leads to a non-zero scour depth at e/y

= 0. This developmental trend from type-a, type-b and type-c to part of type-d is based on reasonable assumptions and analysis and is represented by the dashed curve on the left side of Fig. 19. The fully submerged cases are investigated both in the present study and in the aforementioned literature. The scour depth exhibits an increasing trend with rising e/y value in the small obstacle elevation region, followed by a decreasing trend in the high e/y value region. Under partially submerged conditions, the scour depth decreases rapidly as e/y increases, eventually reaching zero at $e/y = 1$, where no scour occurs. At constant obstacle elevation and flow depth, increasing obstacle height (h/y) will increase the scour depth, while varying obstacle length (l/y) in a reasonable range shows limited influence on the scour depth. Additionally, at constant obstacle elevation and flow depth, increasing the obstacle height does not affect the scour depth once the obstacle becomes partially submerged.

The variation trend of scour depth beneath porous obstacles is generally similar to that observed for solid obstacles. However, within the porosity range of 25 %< φ <75 %, porous obstacles show smaller scour depths. Outside this range, porous obstacles either behave similarly to solid ones (e.g., φ<25 %) or become too permeable (e.g., φ>75 %) to generate scour.

Establishing the fundamental relationship between scour depth and obstacle elevation provides a valuable reference for assessing the scour risk beneath inundated obstacles, such as bridges, floating wetlands, and accumulated debris.

Equations for predicting pressure scour beneath bridge inundation conditions (i.e., solid obstacles) have been widely studied. However, predictive equations for pressure scour induced by porous obstacles remain largely underexplored. Building upon the empirical equation proposed by Melville (2014), the present study develops a predictive equation for pressure scour beneath porous obstacles. In Melville (2014), pressure scour depth is treated as being primarily governed by flow intensity u/u_c, expressed as follows:

$$\begin{cases} \dfrac{y_s}{y_0} = 0.75\left(\dfrac{u}{u_c} - 0.4\right), 0.4 < \dfrac{u}{u_c} < 1 \\ \dfrac{y_s}{y_0} = 0.45, 1 < \dfrac{u}{u_c} < 2.5 \end{cases} \quad (48)$$

Based on this, we further incorporate the influence of e/y and porosity into Eq. (48):

$$\frac{y_s}{y_0} = f\left(\frac{u}{u_c}, \frac{e}{y}, \varphi\right) = \alpha_s K_u K_e K_\varphi \quad (49)$$

where K_u is the same as in Eq. (48) from Melville (2014); K_e and K_φ are determined through curve fitting using the simulated data from the present study; α_s is safety adjustment coefficient for scour depth, for which a value of 3.6 is adapted in the present study. It is worth noting that reducing α_s from the current value of 3.6 to 3.0 improves the fitting performance. However, this reduction causes a larger portion of underestimated predictions, which is not sufficiently safe for engineering applications. Therefore, a more conservative value of α_s equal to 3.6 is recommended. Since all varying parameters were expressed in non-dimensional form, the scale effect in prototype applications is expected to be limited.

As shown in Fig. 20(a) and (b), the black lines are the fitted curves using varying obstacle elevation and porosity, respectively. Although the effect of obstacle height (h/y) is not included in the present equation due to the limited dataset, it should be explicitly considered, as increased obstacle height can induce greater scour depth and cause potential safety concerns. To address this, the blue lines showing the fitted envelope curves for obstacles with increased height are added, generated from the simulated data of solid obstacles. The fitted envelope curves for K_e and K_φ are expressed as follows:

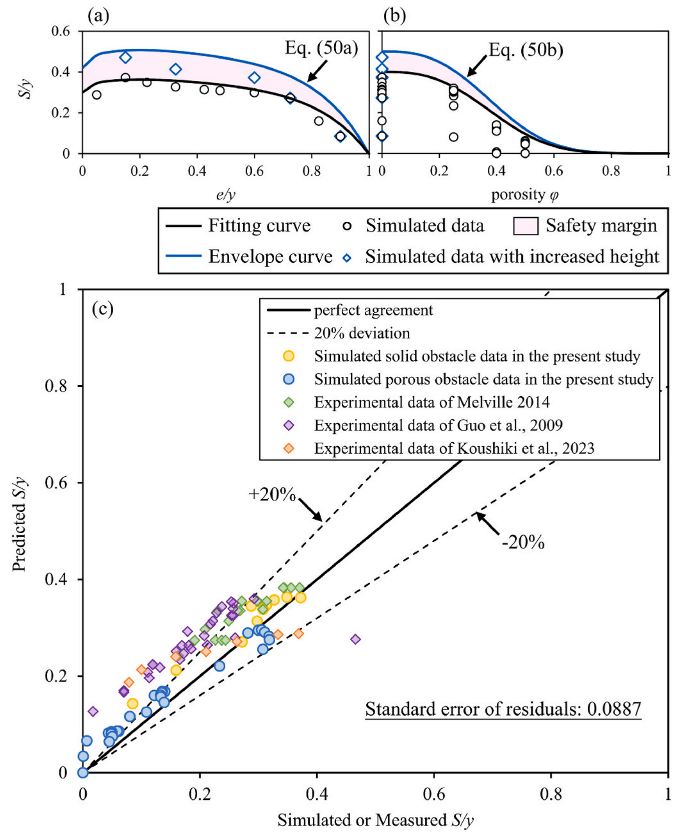

Fig. 20. Newly proposed empirical equation for predicting scour beneath porous obstacles: (a) curve fitting for the effect of e/y; (b) curve fitting for the effect of φ; (c) performance of the newly proposed empirical equation.

$$K_e = 0.42\left(1 - (e/y)^6\right) + 0.308(e/y)^{0.5}(1 - e/y)^2, 0 < \frac{e}{y} < 1 \quad (50a)$$

$$K_\varphi = 0.5e^{-10\varphi^{2.8}} \quad (50b)$$

Fig. 20(c) illustrates the performance of the newly proposed equation for predicting scour beneath porous obstacles. The solid line represents perfect agreement between the simulated (or experimental) and predicted results, while the dashed lines denote a 20 % deviation from the perfect agreement line.

As shown in Fig. 20(c), the results for porous obstacles (blue circles) are mostly located along the perfect agreement line and the standard error of residual is about 0.0887, demonstrating the capability of the proposed equation to predict pressure scour depth across a range of porosities, e.g., 25 % < φ<75 %. Moreover, this empirical equation is also applicable to scour beneath solid obstacles. The predicted values align well with the simulated scour depths beneath solid obstacles obtained in the present study, as well as with the data reported by Melville (2014). However, the predicted results for Guo et al. (2009) and Koushiki et al. (2023) are mostly conservative. Nonetheless, such conservative results are acceptable in practical engineering applications, where conservative predictions provide a necessary safety margin. An exception is observed in the data extracted from Guo et al. (2009), shown as the purple diamond below the −20 % dashed line in Fig. 20(c). This underestimation may be attributed to variations in bridge deck types used in some of the experiments by Guo et al. (2009), specifically a thin bridge deck with six girders. Such differences in bridge deck configuration may lead to shifts in the peak location of the maximum scour depth relative to e/y. As a result, their experiments detected the highest scour depth at a small e/y regime, i.e., $e/y = 0.1$. The effect of bridge deck type needs further investigation. Generally, the newly proposed empirical equation is accurate and conservative for predicting vertical contraction scour depth, especially for porous obstacles.

The limitations and further study concentrations are also worthy of consideration. Firstly, a slip top boundary is used in this study without modelling a varying free surface. This may raise concerns that potential water surface variations, particularly in partially submerged cases, are not fully represented. However, in the present study, the flow depth is sufficiently large such that minor the free surface fluctuations would have minimal influence on the near bed hydrodynamics. Consequently, the impact of this assumption on the simulated scour depth is insignificant. Nevertheless, careful validation is necessary when applying a slip top boundary in shallow water conditions. Additionally, future experimental and numerical studies on scour beneath solid and porous obstacles with a broader range of varying porosity, elevation, length, and height are required to validate the coefficients of the proposed empirical equation and to extensively incorporate the influence of obstacle height and length. Furthermore, only steady current conditions are investigated in the present study. More complex hydrodynamic conditions are possible in real world scenarios and essential for further research, e.g., vertical contraction scour beneath solid and porous obstacles under tidal current with reversing directions, wave conditions, and combined wave and current conditions. Especially in the presence of waves, certain types of floating obstacles may move in response to wave propagation, e.g., a floating breakwater (Cebada-Relea et al., 2022; Zhang et al., 2025), resulting in different scour processes.

Additionally, the present study concentrated more on local scour depth, its interactions with other coastal and hydraulic infrastructures and morphology processes are necessary to be studied in the future. For example, when undercutting occurs beneath certain scour protection methods (Tang et al., 2023, 2024), the resulting process can be considered a combination of pier scour and vertical contraction scour. In the present study, the porous obstacle is simplified; therefore, future research should consider simulating scour beneath porous structures with high-fidelity geometries, to better capture the complexity of real-world scenarios.

8. Conclusions

Numerical investigations and theoretical analysis are conducted in the present study on vertical contraction scour beneath solid and porous obstacles. The numerical model is validated by experimental data. Scour profiles, the evolution of maximum scour locations, and maximum scour depths under varying obstacle elevations and porosities are systematically analyzed. Based on the findings, the following conclusions can be obtained from the present study.

1. Obstacle elevation (e/y) strongly influences maximum scour depth through its effect on flow rate distribution and bed shear stress. For solid obstacles under fully submerged conditions, scour depth increases with e/y at low ranges, then decreases as e/y continues to increase, producing a peak (e.g., at $e/y = 0.15$ in the present study) between small and high e/y regimes. With a further increase in e/y and once the obstacle becomes partially submerged, the scour depth decreases rapidly. Porous obstacles show a similar overall trend but a weaker turning point as flow is allowed to pass through it and the effect of e/y is less pronounced.
2. Obstacle porosity plays an important role in determining both the maximum scour depth and its location. For porous obstacles, the maximum scour occurs near the downstream edge of the obstacle, while for solid obstacles, it is typically concentrated near the center. This difference provides valuable guidance for designing scour protection, particularly in reinforcing the most vulnerable regions. Increasing obstacle porosity results in a notable reduction in maximum scour depth. The present investigation reveals that when $\varphi = 25$ %, the maximum scour depth is comparable to that of solid obstacles. However, as porosity increases to $\varphi = 40$ % and $\varphi = 50$ %, the maximum scour depth decreases significantly. For instance, compared to $\varphi = 25$ %, the maximum scour depth reduced by about 62 % when $\varphi = 40$ %; and when $\varphi = 50$ %, the maximum scour depth reduced by about 86 %.
3. Under constant flow depth and obstacle elevation, an increase in obstacle height (h/y) leads to a greater maximum scour depth. Conversely, variations in obstacle length (l/y) have a limited impact on the maximum scour depth.
4. Theoretical analysis can well capture the flow rate distribution, average velocity and its corresponding bed shear stress at the initial stage, showing good agreements between theoretical and numerical results for both solid and porous obstacles.
5. Basic relationship between obstacle elevation and maximum vertical contraction scour depth for both solid and porous obstacles are presented. These findings provide a comprehensive summary of the key parameters influencing vertical contraction scour and serve as a valuable reference for the future development of predictive equations.
6. A new empirical equation for predicting scour depth beneath porous obstacles is proposed. This equation demonstrates good agreement with the simulated or measured results, demonstrating its capability for practical applications.

CRediT authorship contribution statement

Zihao Tang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Yuzhu Pearl Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The corresponding author: Given her role as Associate Editor of Coastal Engineering, Yuzhu Pearl Li had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another editor of the journal. The remaining authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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